# Algorithms and Analysis

## Lesson 2: *Know How Long A Program Takes*

*TSP, Sorting, time complexity, Big-Theta, Big-O, Big-Omega*



# Outline

1. **TSP**
2. Sorting
3. Big O

# Travelling Salesperson Problem

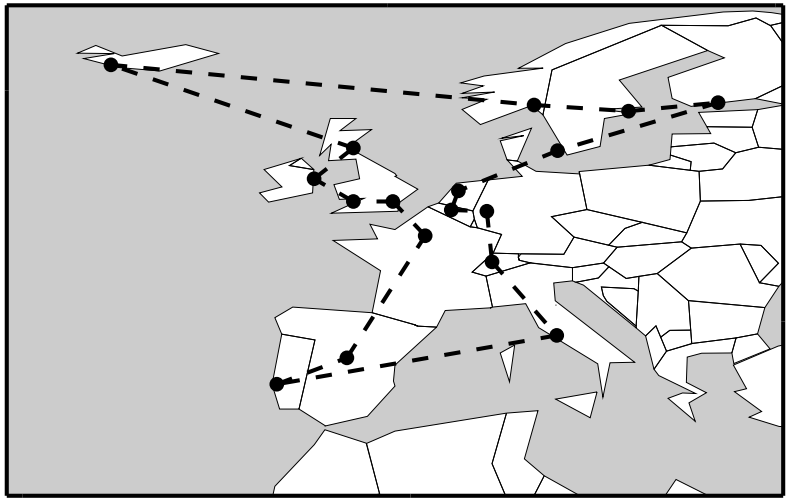
- Given a set of cities
- A table of distances between cities
- Find the shortest tour which goes through each city and returns to the start



# Example of Distance Table

| | Lon | Car | Dub | Edin | Reyk | Oslo | Sto | Hel | Cop | Amst | Bru | Bonn | Bern | Rome | Lisb | Madr | Par |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| London | 0 | 223 | 470 | 538 | 1896 | 1151 | 1426 | 1816 | 950 | 349 | 312 | 503 | 743 | 1429 | 1587 | 1265 | 337 |
| Cardiff | 223 | 0 | 290 | 495 | 1777 | 1277 | 1589 | 1985 | 1139 | 564 | 533 | 725 | 927 | 1600 | 1492 | 1233 | 492 |
| Dublin | 470 | 290 | 0 | 350 | 1497 | 1267 | 1628 | 2026 | 1239 | 756 | 775 | 956 | 1207 | 1886 | 1638 | 1449 | 777 |
| Edinburgh | 538 | 495 | 350 | 0 | 1374 | 933 | 1314 | 1708 | 984 | 662 | 758 | 896 | 1243 | 1931 | 1964 | 1728 | 872 |
| Reykjavik | 1896 | 1777 | 1497 | 1374 | 0 | 1746 | 2134 | 2418 | 2104 | 2020 | 2130 | 2255 | 2617 | 3304 | 2949 | 2892 | 2232 |
| Oslo | 1151 | 1277 | 1267 | 933 | 1746 | 0 | 416 | 788 | 481 | 917 | 1088 | 1048 | 1459 | 2011 | 2739 | 2390 | 1343 |
| Stockholm | 1426 | 1589 | 1628 | 1314 | 2134 | 416 | 0 | 398 | 518 | 1126 | 1281 | 1181 | 1542 | 1978 | 2987 | 2593 | 1543 |
| Helsinki | 1816 | 1985 | 2026 | 1708 | 2418 | 788 | 398 | 0 | 881 | 1504 | 1650 | 1530 | 1856 | 2203 | 3360 | 2950 | 1910 |
| Copenhagen | 950 | 1139 | 1239 | 984 | 2104 | 481 | 518 | 881 | 0 | 625 | 769 | 662 | 1036 | 1538 | 2479 | 2076 | 1030 |
| Amsterdam | 349 | 564 | 756 | 662 | 2020 | 917 | 1126 | 1504 | 625 | 0 | 173 | 235 | 629 | 1296 | 1860 | 1480 | 428 |
| Brussels | 312 | 533 | 775 | 758 | 2130 | 1088 | 1281 | 1650 | 769 | 173 | 0 | 194 | 489 | 1174 | 1710 | 1315 | 262 |
| Bonn | 503 | 725 | 956 | 896 | 2255 | 1048 | 1181 | 1530 | 662 | 235 | 194 | 0 | 422 | 1067 | 1843 | 1420 | 400 |
| Bern | 743 | 927 | 1207 | 1243 | 2617 | 1459 | 1542 | 1856 | 1036 | 629 | 489 | 422 | 0 | 689 | 1630 | 1156 | 440 |
| Rome | 1429 | 1600 | 1886 | 1931 | 3304 | 2011 | 1978 | 2203 | 1538 | 1296 | 1174 | 1067 | 689 | 0 | 1862 | 1365 | 1109 |
| Lisbon | 1587 | 1492 | 1638 | 1964 | 2949 | 2739 | 2987 | 3360 | 2479 | 1860 | 1710 | 1843 | 1630 | 1862 | 0 | 500 | 1452 |
| Madrid | 1265 | 1233 | 1449 | 1728 | 2892 | 2390 | 2593 | 2950 | 2076 | 1480 | 1315 | 1420 | 1156 | 1365 | 500 | 0 | 1054 |
| Paris | 337 | 492 | 777 | 872 | 2232 | 1343 | 1543 | 1910 | 1030 | 428 | 262 | 400 | 440 | 1109 | 1452 | 1054 | 0 |

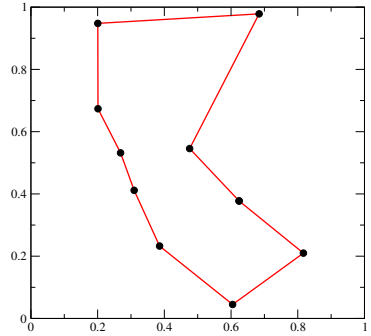
# Example Tour





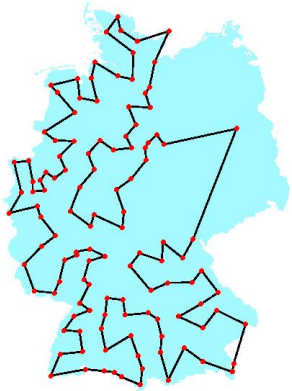
# Brute Force

- I wrote a program to solve TSP by enumerating every path and finding the shortest
- I checked that it worked on some problems with 10 cities
- It takes just under half a second to solve this problem
- I set the program running on a 100 city problem—**How long will it take to finish?**





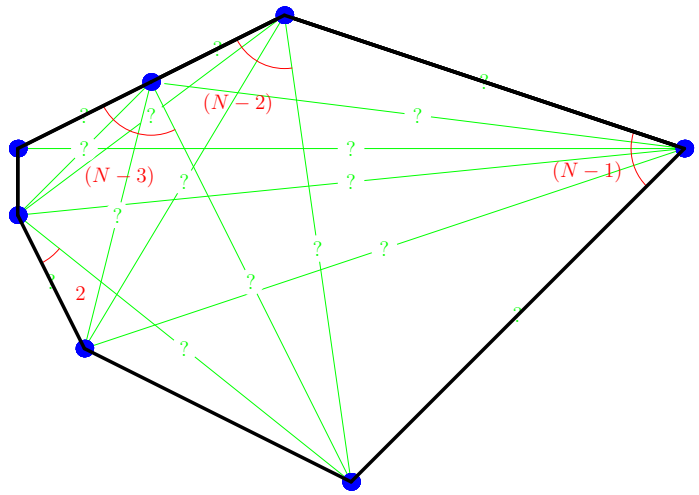

# How Many Possible Tours Are There?



- For 100 cities how many possible tours are there?
- It doesn't matter where we start
- Starting from Berlin there are 99 cites we can try next



# Counting Tours




Number of tours = $(N-1) \times (N-2) \times (N-3) \times \cdots 2 \times 1 = (N-1)!$

## How Long Does It Take?

- The direction we go in is irrelevant
- Total number of tours is $99!/2$
- **Any more guesses how long it will take?**



## How Big is 99 Factorial?

- $99! = 99 \cdot 98 \cdot 97 \cdots 2 \cdot 1 = ?$
- Upper bound

$$\begin{aligned} 99! &= 99 \cdot 98 \cdot 97 \cdots 2 \cdot 1 \\ 99! &< 99 \cdot 99 \cdot 99 \cdots 99 \cdot 99 = 99^{99} \end{aligned}$$

- Lower bound

$$\begin{aligned} 99! &= 99 \cdot 98 \cdot 97 \cdots 50 \cdot 49 \cdots 2 \cdot 1 \\ 99! &> 50 \cdot 50 \cdot 50 \cdots 50 \cdot 1 \cdots 1 \cdot 1 = 50^{50} \end{aligned}$$



## How Long Does It Take?

- For $N > 1$

$$\left(\frac{N}{2}\right)^{N/2} < N! < N^N$$

- $99!/2 = 4.666 \times 10^{155}$
- How long does it take to search all possible tours?
  - We computed about $200\,000$ tours in half a second
  - $3.15 \times 10^7$sec $= 1$ year
  - Age of Universe $\approx 15$ billion years

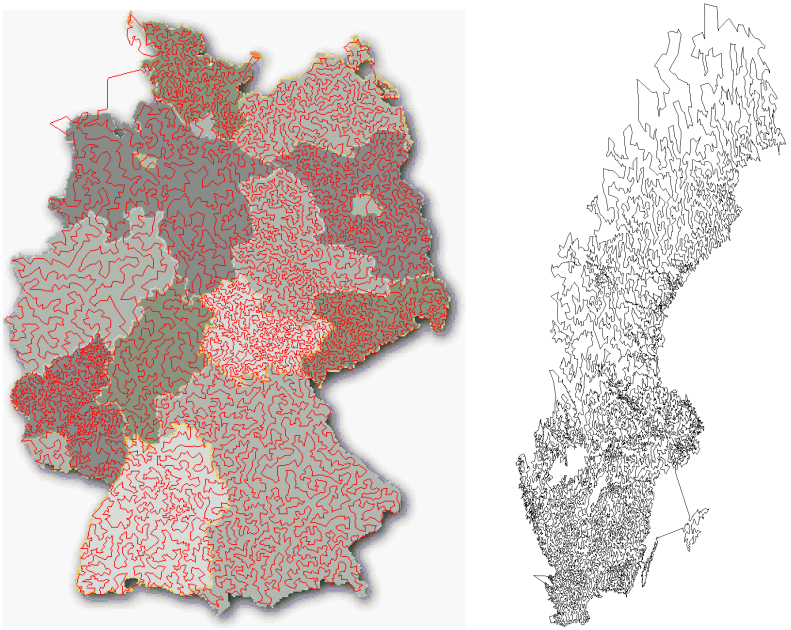


## Answer

- $2.72 \times 10^{132}$ ages of the universe!
- Incidental

$$\begin{aligned} 99!/2 = {} & 46663107721972076340849619428133350 \\ & 24535798413219081073429648194760879 \\ & 99966149578044707319880782591431268 \\ & 48960413611879125592605458432000000 \\ & 0000000000000000 \end{aligned}$$

## Record TSP Solved—$15\,112$ and $24\,978$ Cities





## In Case You're Curious

- Number of tours: $15111!/2 = 7.3 \times 10^{56592}$
- Current record $24\,978$ cities with $1.9 \times 10^{98992}$ tours
- The algorithm for finding the optimum path does not look at every possible path
- If your interested look for the TSP homepage on the web `http://www.math.uwaterloo.ca/tsp/`



## Lessons

- Even relatively small problems can take you an astronomical time to solve using simple algorithms
- As a professional programmer you need to have an estimate for how long an algorithm takes—otherwise you can look silly
- For the 100 city problem, if
  - I had $10^{87}$ cores, one for every particle in the Universe
  - I could compute a tour distance in $3 \times 10^{-24}$ seconds, the time it takes light to cross a proton
  - It would still take $10^{39}\times$ the age of the universe
- Smart algorithms can make a much larger difference than fast computers!



## Outline

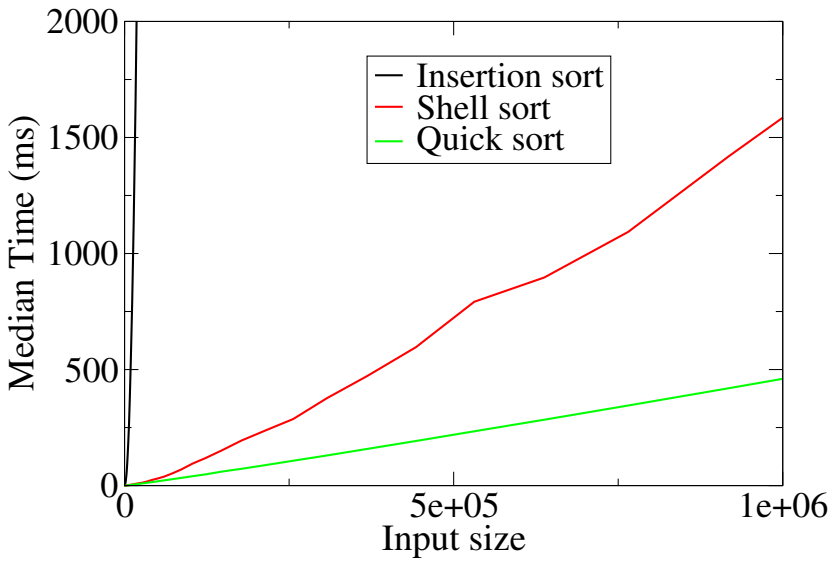
1. TSP
2. **Sorting**
3. Big O

## Sort

- Comparison between common sort algorithms
  - Insertion sort—an easy algorithm to code
  - Shell sort—invented in 1959 by Donald Shell
  - Quick sort—invented in 1961 by Tony Hoare
- These take an array of numbers and returns a sorted array
- Sort is very commonly used algorithm so you care about how long it takes



## Empirical Run Times






## Lessons

- There is a right and wrong way to do easy problems
- You only really care when you are dealing with large inputs
- Good algorithms are difficult to come up with, but they exist
- We would like to quantify the performance of an algorithm—how much better is quick sort than insertion sort?



## Outline

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## Estimating Run Times

- We would like to estimate the run times of algorithms
- This depends on the hardware (how fast is your computer)
- We could count the number of elementary operations, but
  - ⋆ different machines have different elementary operations
  - ⋆ many algorithms use complex functions such as `sqrt` (matrix inversion using Cholesky decomposition) or `sin` and `cos` (FFT)
  - ⋆ would need to count memory accesses which you shouldn't need to think about
  - ⋆ code after compiling can be very different from code before compiling



## Engineering Solution

- Compute the **asymptotic leading functional behaviour**
- Lets take that statement to pieces
- Suppose we have an algorithm that takes $4n^2 + 12n + 199$ operations (clock cycles)
  - ⋆ **asymptotic**: what happens when $n$ becomes very large
  - ⋆ **leading**: ignore the $12n + 199$ part as it is dominated by $4n^2$ (i.e. for large enough $n$ we have $4n^2 \gg 12n + 199$)
  - ⋆ **functional behaviour**: ignore the constant $4$
- We call this an order $n^2$, or quadratic time, algorithm
- We can write this in 'Big-Theta' notation as $\Theta(n^2)$
- This notion of 'run time' is known as **time complexity**



## Advantages of Big-Theta Notation

- Doesn't depend on what computer we are running
- Don't need to know how many elementary operations are required for a non-elementary operation
- Can estimate run times by measuring run time on a small problem
  - ⋆ If I have a $\Theta(n^2)$ algorithm
  - ⋆ It takes $x$ seconds on an input of $100$
  - ⋆ It will take about $\frac{x \times n^2}{100^2}$ seconds on a problem of size $n$
    ($T(100) \approx c\,100^2 = x$ therefore $c = x/100^2$
    thus $T(n) = c\,n^2 = x\,n^2/100^2$)



## Counting Instructions

- Big-Theta run times are often easy to calculate
- a $\Theta(n)$ algorithm

```
// define stuff
for(int i=0; i<n; i++)  {
  // do something
}
// clean up
```

- a $\Theta(n^2)$ algorithm

```
// define stuff
for(int i=0; i<n; i++)  {
  // do something
  for (int j=0; j<n; j++) {
    // do other stuff
  }
}
// clean up
```

## Disadvantage with Big-Theta notation

- Can't compare algorithms with the same Big-Theta time complexity
- For small inputs Big-Theta time complexity can be misleading. E.g.
  - ⋆ algorithm A takes $n^3 + 2n^2 + 5$ operations
  - ⋆ algorithm B takes $20n^2 + 100$ operations
  - ⋆ algorithm A is $\Theta(n^3)$ and algorithm B is $\Theta(n^2)$
  - ⋆ algorithm A is faster than algorithm B for $n < 18$

  but who cares?
- In some cases Big-Theta time complexity is hard to compute



## Not So Sure

- Some algorithms are harder to compute

```
// define stuff
for(int i=0; i<n; i++)  {
  // do something
  if (/* some condition */) {
    for (int j=0; j<n; j++) {
      // do other stuff
    }
  }
}
// clean up
```

- Time complexity now depends on the `if` statement
- If the condition is often satisfied we have a $\Theta(n^2)$ algorithm
- If the condition is true only rarely then we have a $\Theta(n)$ algorithm



## Bounds

- To avoid having to think really hard we define upper and lower bounds
- The upper bound we write using **big-O** notation
  - ⋆ The above algorithm is an $O(n^2)$ algorithm
  - ⋆ I.e. it runs in no more than order $n^2$ operations
- The lower bound we write using **big-Omega** notation
  - ⋆ The above algorithm is a $\Omega(n)$ algorithm
  - ⋆ I.e. it runs in no less than order $n$ operations



## Precise Definitions of $O(n)$

- An algorithm that runs in $f(n)$ operations is $O(g(n))$ if

$$\lim_{n\to\infty} \frac{f(n)}{g(n)} = c \qquad \text{where } c \text{ is a constant (could be zero)}$$

- E.g.. $f(n) = 3\,n^2 + 2\,n + 12$

$$\lim_{n\to\infty} \frac{f(n)}{g(n)} = \lim_{n\to\infty} \frac{3\,n^2 + 2\,n + 12}{n^2} = 3 \quad \Rightarrow 3\,n^2 + 2\,n + 12 = O(n^2)$$

$$\lim_{n\to\infty} \frac{f(n)}{g(n)} = \lim_{n\to\infty} \frac{3\,n^2 + 2\,n + 12}{n^3} = 0 \quad \Rightarrow 3\,n^2 + 2\,n + 12 = O(n^3)$$

$$\lim_{n\to\infty} \frac{f(n)}{g(n)} = \lim_{n\to\infty} \frac{3\,n^2 + 2\,n + 12}{n} = \infty \quad \Rightarrow 3\,n^2 + 2\,n + 12 \neq O(n)$$

## Lower Bound Definition

- An algorithm that runs in $f(n)$ operations is $\Omega(g(n))$ if

$$\lim_{n\to\infty} \frac{g(n)}{f(n)} = c \qquad \text{where } c \text{ is a constant (could be zero)}$$

- E.g. $f(n) = 3\,n^2 + 2\,n + 12$

$$\lim_{n\to\infty} \frac{g(n)}{f(n)} = \lim_{n\to\infty} \frac{n^2}{3\,n^2 + 2\,n + 12} = \frac{1}{3} \Rightarrow 3\,n^2 + 2\,n + 12 = \Omega(n^2)$$

$$\lim_{n\to\infty} \frac{g(n)}{f(n)} = \lim_{n\to\infty} \frac{n^3}{3\,n^2 + 2\,n + 12} = \infty \Rightarrow 3\,n^2 + 2\,n + 12 \neq \Omega(n^3)$$

$$\lim_{n\to\infty} \frac{g(n)}{f(n)} = \lim_{n\to\infty} \frac{n}{3\,n^2 + 2\,n + 12} = 0 \Rightarrow 3\,n^2 + 2\,n + 12 = \Omega(n)$$



## Big-Theta

- An algorithm that runs in $f(n)$ operations is $\Theta(g(n))$ if

$$\lim_{n\to\infty} \frac{g(n)}{f(n)} = c \qquad \text{where } c \text{ is a non-zero constant}$$

- That is, $f(n) = \Theta(g(n))$ if

$$f(n) = O(g(n)) \quad \text{and} \quad f(n) = \Omega(g(n))$$

- I.e. the lower bound is identical to the upper bound
- Often the most straightforward way of obtaining big-Theta is to show the upper and lower bounds are the same



## Use and Misuse

- Note: big-O notation is most commonly used
- often people say they have a $O(n^2)$ when in fact they mean they have a $\Theta(n^2)$ algorithm (a much stronger result)
- Note that an $O(n^2)$ algorithm is also a $O(n^3)$ algorithm
- Strictly a $O(n^2)$ algorithm **may not** be faster than a $O(n^3)$ algorithm when $n$ becomes larger
- A $\Theta(n^2)$ algorithm **will** be faster than a $\Theta(n^3)$ algorithm when $n$ becomes larger



## Lessons to Learn

- Run times (computational time complexity) matters
- Choosing an algorithm with the best time complexity is important
- Understand the meaning of big-Theta, big-O and big-Omega
- Know how to estimate time complexity for simple algorithms (loop counting)